Supplementary Material

QSMEF: A Methodological Framework for Functional Analysis of Quantum Implementations

Additional Application Case: QAOA–MaxCut

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September 2026

Abstract

This supplementary document complements the paper *QSMEF: A Methodological Framework for Functional Analysis of Quantum Implementations* with an additional application of QSMEF to the Quantum Approximate Optimization Algorithm (QAOA) for MaxCut.

The experiment considers a six-vertex cycle and a QAOA circuit with $p = 2$ using a fixed parameter set. The analysis first verifies the behavior of the complete QAOA instance and then applies QSMEF to a functional decomposition consisting of four cost and mixer blocks.

For each coalition, the experiment preserves the original operational order, replaces absent functional components with the identity operator, and keeps the graph, initial state, parameters, and selected observable fixed. QSMEF evaluates all 16 coalitions and uses Shapley values to attribute the change in the expected value of the observable among the four functional components.

The results provide an additional application of QSMEF in a computational setting distinct from the SKW quantum search analyzed in the companion paper. They also illustrate that Shapley contributions represent average marginal effects across coalition contexts rather than isolated effects of individual circuit blocks.

1 Introduction

The companion paper introduces QSMEF and evaluates it through the Shenvi–Kempe–Whaley (SKW) coined quantum-walk search algorithm on the hypercube [3]. The SKW case analyzes the functional contributions of the oracle, the Grover coin operator, and the flip-flop shift operator with respect to a selected Hermitian observable.

This supplementary document extends the empirical evaluation to a different computational structure: the Quantum Approximate Optimization Algorithm (QAOA) [1] applied to MaxCut. The analysis uses a fixed QAOA instance based on the formulation and parameters reported by Wang et al. [4]. Its objective is not to reproduce their complete optimization procedure, but to use a published parameterization as a reference instance for functional attribution.

The analysis first verifies that the complete implementation reproduces the reference QAOA behavior reported by Wang et al. [4]. QSMEF then decomposes the parameterized circuit into four functional blocks and evaluates all their coalitions while keeping the graph, initial state, parameters, observable, and operational order fixed.

This experiment provides an additional application case for QSMEF and examines its use for quantifying functional contributions in an algorithmic structure distinct from coined quantum-walk search.

The remainder of this supplementary document is organized as follows. Section 2 introduces the QAOA–MaxCut model and the problem instance considered in the experiment. Section 3 defines the Hermitian observable and the functional metric used in the analysis. Section 4 presents the functional decomposition adopted by QSMEF and the fixed QAOA parameterization. Section 5 verifies the behavior of the complete implementation against the reference QAOA result. Section 6 defines the cooperative game used for functional attribution, and Section 7 presents the resulting Shapley values. Section 8 reports the results for all coalition configurations, while Section 9 describes the numerical verification procedures. Finally, Section 10 discusses the results and limitations of the experiment, and Section 11 presents the main conclusions.

2 QAOA–MaxCut Model

2.1 Problem Instance

The experiment considers MaxCut on a cycle of six vertices, following the QAOA–MaxCut setting analyzed by Wang et al. [4]:

$$V = \{0, 1, 2, 3, 4, 5\}, \tag{1}$$

with edge set

$$E = \{(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 0)\}. \tag{2}$$

Each qubit represents one vertex. A binary string $z \in \{0, 1\}^6$ assigns each vertex to one of two subsets and defines the number of cut edges as

$$c(z) = \sum_{(i,j)\in E} \mathbf{1}[z_i \neq z_j]. \tag{3}$$

The maximum cut of the cycle C_6 contains six edges. The strings 010101 and 101010 represent equivalent partitions that cut all six edges.

The Hilbert space of the experiment is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 6}, \qquad \dim \mathcal{H} = 64. \tag{4}$$

Following the standard QAOA construction [1], the analysis uses the initial state

$$|\psi_0\rangle = |+\rangle^{\otimes 6}, \qquad \rho_0 = |\psi_0\rangle\langle\psi_0|. \tag{5}$$

The Hadamard gates that prepare this state remain fixed and do not belong to the set of attributable functional components.

3 Observable and Functional Metric

The experiment follows the minimization convention used in the reference QAOA formulation by Wang et al. [4] and adopts

$$H_C = \sum_{j=0}^{5} Z_j Z_{(j+1) \bmod 6} \tag{6}$$

as the Hermitian observable.

For each coalition C, QSMEF evaluates

$$M_{H_C}(\rho_C) = \mathrm{Tr}(H_C \rho_C) = F(C). \tag{7}$$

Under this convention, lower values of $F(C)$ favor the MaxCut objective.

For interpretation in terms of the expected number of cut edges, we also define

$$H_{\text{cut}} = \frac{6I_{64} - H_C}{2}, \tag{8}$$

which gives

$$K(C) = \mathrm{Tr}(H_{\text{cut}} \rho_C) = \frac{6 - F(C)}{2}. \tag{9}$$

Thus, minimizing the expected value of H_C is equivalent to maximizing the expected number of cut edges. QSMEF uses H_C as the observable for the cooperative game, while H_{cut} provides a complementary interpretation of the resulting values.

4 QSMEF Functional Decomposition

For depth $p = 2$, the QAOA circuit contains four functional blocks:

$$B_{\text{MaxCut}} = \{C_1, M_1, C_2, M_2\}. \tag{10}$$

Following the standard QAOA construction [1], each QAOA layer contains one cost block and one mixer block:

$$C_k = e^{-i\gamma_k H_C}, \qquad M_k = e^{-i\beta_k H_{\text{mix}}}, \qquad k = 1, 2, \tag{11}$$

where

$$H_{\text{mix}} = \sum_{j=0}^{5} X_j. \tag{12}$$

Using the conventions

$$R_{ZZ}(\theta) = e^{-i\theta Z \otimes Z/2}, \qquad R_x(\theta) = e^{-i\theta X/2}, \tag{13}$$

the functional blocks take the form

$$C_k = \prod_{(i,j)\in E} R_{ZZ}^{(i,j)}(2\gamma_k), \qquad M_k = \bigotimes_{j=0}^{5} R_x(2\beta_k). \tag{14}$$

The experiment uses the second parameter set reported for $p = 2$ by Wang et al. [4]:

$$(\gamma_1, \beta_1, \gamma_2, \beta_2) = \pi(0.2052,\ 0.1026,\ 0.3974,\ 0.2948). \tag{15}$$

The analysis retains the four published decimal places and does not re-optimize the parameters. For every coalition $C \subseteq B_{\mathrm{MaxCut}}$, QSMEF preserves the original circuit order

$$C_1 \to M_1 \to C_2 \to M_2. \tag{16}$$

If a component does not belong to a coalition, QSMEF replaces it with the identity operator at its original position. Therefore, all coalitions use the same graph, initial state, parameter values, observable, and operational structure.

Figure 1 shows the $p = 2$ QAOA circuit and the functional decomposition adopted for the QSMEF analysis.

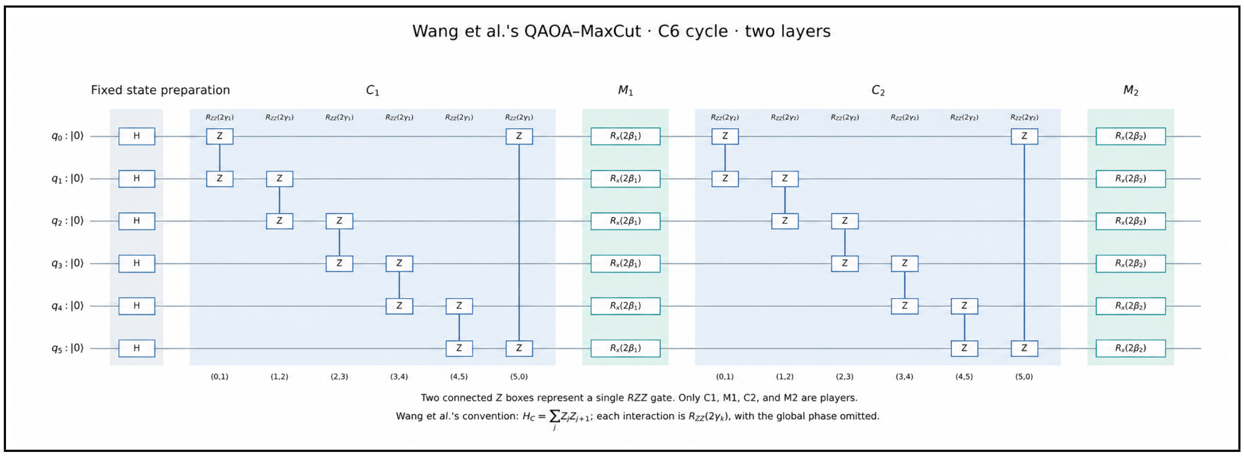

Figure 1: QAOA–MaxCut circuit for the $p = 2$ instance. The functional decomposition used by QSMEF consists of the four blocks C_1, M_1, C_2, and M_2.

5 Reproduction of the Reference QAOA Result

Before computing the functional attribution, the experiment verifies the behavior of the complete QAOA implementation.

The simulation uses statevectors with 64 amplitudes and 64×64 matrices under ideal conditions, without noise or finite sampling.

For the empty coalition,

$$F(\emptyset) = 0, \qquad K(\emptyset) = 3. \tag{17}$$

For the complete circuit, the simulation produces

$$F(B_{\mathrm{MaxCut}}) \simeq -3.9999997087, \tag{18}$$

and therefore

$$K(B_{\mathrm{MaxCut}}) \simeq 4.9999998544. \tag{19}$$

The normalized objective value is

$$\frac{F(B_{\mathrm{MaxCut}})}{6} \simeq -0.6666666181, \tag{20}$$

and the corresponding approximation ratio is

$$r = \frac{K(B_{\mathrm{MaxCut}})}{6} \simeq 0.8333333091. \tag{21}$$

These values agree with the reference values

$$\frac{F}{n} = -\frac{2}{3} \tag{22}$$

and

$$r = \frac{5}{6}, \tag{23}$$

reported by Wang et al. [4] for the $p = 2$ case.

The absolute difference in F/n is approximately

$$4.85 \times 10^{-8}, \tag{24}$$

which is consistent with the precision of the published parameters. The experiment uses a reproduction tolerance of 10^{-6}.

The expected cut value close to five does not imply that the circuit always produces a maximum cut. The probability of measuring one of the assignments that cuts all six edges is approximately

$$0.536312. \tag{25}$$

This probability represents a different property from both the expected cut value $K(B_{\mathrm{MaxCut}})$ and the approximation ratio r.

6 QSMEF Cooperative Game

Once the experiment fixes the QAOA instance, QSMEF evaluates all

$$2^4 = 16 \tag{26}$$

possible coalitions without modifying the parameters or the original order of operations.

The characteristic function uses the expected value of H_C:

$$v(C) = F(C) - F(\emptyset), \qquad v(\emptyset) = 0. \tag{27}$$

Because the analysis follows a minimization convention, a negative Shapley contribution represents an average marginal effect that favors a reduction in $F(C)$.

For each functional block $b \in B_{\mathrm{MaxCut}}$, QSMEF computes the Shapley value [2] as

$$\phi_b = \sum_{C \subseteq B_{\mathrm{MaxCut}} \setminus \{b\}} \frac{|C|!(3 - |C|)!}{4!} \left[F(C \cup \{b\}) - F(C) \right]. \tag{28}$$

7 Functional Attribution Results

Table 1 presents the Shapley values obtained for the four functional blocks.

Table 1: Shapley contributions to the change in the expected value of H_C for the QAOA–MaxCut instance.

Block	$\boldsymbol{\phi_b}$
C_1	-1.583908073
M_1	-0.416091782
C_2	-0.416091782
M_2	-1.583908073
Sum	$\mathbf{-3.999999709}$

The efficiency property gives

$$\sum_{b \in B_{\text{MaxCut}}} \phi_b = v(B_{\text{MaxCut}}) = F(B_{\text{MaxCut}}) - F(\emptyset) \simeq -3.9999997087. \tag{29}$$

Thus, the individual Shapley contributions sum to the total change in the selected functional metric relative to the empty-coalition baseline.

For this instance, C_1 and M_2 receive contributions with larger magnitudes than M_1 and C_2. This result does not imply that the isolated presence of these blocks necessarily produces the same effect.

For example, C_1 acting alone gives

$$F(\{C_1\}) = 0, \tag{30}$$

although its Shapley contribution is nonzero. Its contribution results from the weighted average of its marginal effects across the different coalition contexts.

Similarly, the coalition

$$\{C_1, M_1\} \tag{31}$$

produces approximately

$$F(\{C_1, M_1\}) \simeq 1.538095, \tag{32}$$

even though both components have negative Shapley contributions in the complete cooperative game.

These results illustrate that a Shapley contribution does not represent the isolated effect of a component. Instead, it summarizes its average marginal effect across the different coalition contexts.

The equalities

$$\phi_{C_1} = \phi_{M_2} \tag{33}$$

and

$$\phi_{M_1} = \phi_{C_2} \tag{34}$$

hold for this particular instance. They do not establish a general property of QAOA cost and mixer blocks.

8 Coalition Results

Table 2 reports the expected value $F(C) = \mathrm{Tr}(H_C \rho_C)$ for all 16 coalitions. Every configuration uses the same Hamiltonian, sign convention, parameters, and operational order. The symbol I denotes the identity operator I_{64}.

Table 2: Results for the 16 coalitions of the QAOA–MaxCut instance.

C_1	M_1	C_2	M_2	$F(C)$	$v(C)$
I	I	I	I	0.000000	0.000000
C_1	I	I	I	0.000000	0.000000
I	M_1	I	I	0.000000	0.000000
I	I	C_2	I	0.000000	0.000000
I	I	I	M_2	0.000000	0.000000
C_1	M_1	I	I	1.538095	1.538095
C_1	I	C_2	I	0.000000	0.000000
I	M_1	C_2	I	0.000000	0.000000
C_1	I	I	M_2	−0.854517	−0.854517
I	M_1	I	M_2	0.000000	0.000000
I	I	C_2	M_2	1.538095	1.538095
C_1	M_1	C_2	I	1.538095	1.538095
C_1	M_1	I	M_2	−1.538095	−1.538095
C_1	I	C_2	M_2	−1.538095	−1.538095
I	M_1	C_2	M_2	1.538095	1.538095
C_1	M_1	C_2	M_2	−4.000000	−4.000000

Since $F(\emptyset) = 0$, the columns $F(C)$ and $v(C)$ coincide numerically in this instance, although they represent conceptually different quantities.

9 Numerical Verification

The experiment performs several numerical checks to verify the implementation.

First, it checks Hermiticity, state normalization, and the unitarity of the operations used in all 16 coalition configurations. It also compares the implemented cost and mixer blocks with an independent construction based on matrix exponentials of the complete Hamiltonians.

The experiment verifies H_{cut} against the classical number of cut edges for all 64 computational-basis strings and compares the expectation values with the corresponding probability-weighted averages.

QSMEF computes the Shapley contributions using two independent procedures: the subset-based Shapley formula and the average over the 24 possible permutations of the four players. The permutations only determine the order in which the marginal contributions enter the Shapley calculation; they do not reorder the operations in the quantum circuit. Every coalition preserves the original operational order.

As an additional check, the analysis computes the attribution of the expected cut value and verifies

$$\phi_b^{\mathrm{cut}} = -\frac{\phi_b}{2}. \tag{35}$$

This check does not modify the cooperative game defined using H_C.

For the reported execution, the residuals of the numerical checks remain below

$$10^{-10}. \tag{36}$$

The efficiency residual is zero within numerical precision. Differences on the order of 10^{-15} result from floating-point rounding and do not by themselves indicate a functional error.

10 Discussion

The QAOA–MaxCut experiment shows that QSMEF can attribute the change in the expected value of a selected observable among the functional blocks of a fixed parameterized circuit when the coalition configurations support a common functional interpretation.

The reproduction of the result reported in the reference QAOA study provides an external reference for the behavior of the complete circuit. QSMEF adds a different level of analysis to this fixed instance: it evaluates partial configurations and distributes the change in $F(C)$ among the four functional blocks through Shapley values.

The resulting contributions do not represent absolute measures of component importance. Each Shapley value summarizes the average marginal effect of a block across the coalition contexts considered by the game. Therefore, a block can receive a contribution that favors the overall minimization objective even when its effect in a particular coalition increases $F(C)$.

For the analyzed instance, C_1 and M_2 have contributions with larger magnitudes than M_1 and C_2. These relationships depend on the graph, parameters, functional decomposition, and selected observable. The experiment therefore does not generalize this attribution pattern to other QAOA configurations.

The experiment uses a small graph, published parameters with limited decimal precision, and ideal simulation conditions. It does not include noise or evaluate scalability. These limitations constrain the scope of the numerical results, but the case provides an additional evaluation of QSMEF in a computational structure different from the SKW quantum search analyzed in the companion paper.

11 Conclusions

This supplementary experiment applied QSMEF to a QAOA–MaxCut instance with $p = 2$ on a six-vertex cycle.

The complete implementation reproduces the reference QAOA behavior, with $F/n \simeq -2/3$ and an approximation ratio close to 5/6. QSMEF then evaluates the 16 coalitions generated by the functional decomposition $\{C_1, M_1, C_2, M_2\}$ while keeping the graph, parameters, initial state, observable, and operational order fixed.

The resulting Shapley values satisfy the efficiency property and attribute the total change in the selected observable-based metric among the four functional blocks. The results also illustrate that these contributions represent average marginal effects across coalition contexts rather than isolated effects of individual blocks.

Together with the SKW case presented in the companion paper, this experiment provides additional evidence that QSMEF can support functional attribution in two distinct quantum algorithmic structures. In contrast, a separate supplementary QPE applicability analysis identifies a configuration in which functional comparability among coalitions does not hold, thereby illustrating an applicability boundary of the framework.

References

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